

Lab Task#5

Sec A

Note: All statements should be commented. Plagiarism will get straight zero.

Use functions (def). Do not use lambda.

Question 1

Approximate $\sqrt{n}$ with the Babylonian method. Start with a guess, then repeatedly update:

$$\text{guess} = (\text{guess} + n / \text{guess}) / 2$$

Write `approximate_sqrt(n, guess=1.0, steps=8)` that performs the update `steps` times and returns the final guess.

Include a docstring. Call it for $n = 49$ with defaults, and again for $n = 2$ with `guess = 1.0` and `steps = 10`.

Print both results.

Question 2

Combinations are given by:

$$C(n, r) = n! / (r! \cdot (n - r)!)$$

Write `factorial(k)` that returns $k!$ using a loop.

Write `combination(n, r)` that returns $C(n, r)$ by calling `factorial`.

Print $C(10, 3)$ and $C(6, 2)$.

Question 3

An arithmetic progression has first term a and common difference d .

nth term: $a_n = a + (n - 1)d$

Sum of n terms: $S_n = n/2 \cdot (2a + (n - 1)d)$

Write `nth_term(a, n, d=1)` and `sum_ap(a, n, d=1)`. Both should return their values.

Use default `d = 1`. Include a short docstring in `sum_ap`.

For `a = 3, n = 10`: print the 10th term and `S_10` once with default `d`, and once with `d = 4`.

Question 4

Positive integers (a, b, c) form a Pythagorean triple if:

$$a^2 + b^2 = c^2$$

Write `is_pythagorean(a, b, c)` that returns True or False.

Then use nested loops with $a \leq b \leq c \leq 30$ to find and print every triple that satisfies the equation.

Print each triple as: `a, b, c`

Also print how many such triples were found.